

PRIYANK AGRAWAL

Seeley W. Mudd Building, 500 W. 120th Street, New York, NY 10027
pa2608@columbia.edu; Website: AgPriyank.github.io; Google Scholar: link

RESEARCH FOCUS

Keywords: Reinforcement learning theory and algorithms (planning and exploration; regret and sample complexity analysis); LLM Post-training (RLHF, RLVR, reward modeling, supervised fine-tuning, alignment); Agentic AI and inference time optimization; mechanism design and strategic behavior.

Research Summary: I develop reinforcement learning algorithms with both theoretical foundations (COLT 2025, ICLR 2026, NeurIPS 2025) and applied impact in LLM post-training. As a student researcher at Meta, I designed RLVR methods for reasoning language models, studying training failure modes and curriculum strategies at scale.

EDUCATION

Columbia University, New York, USA *Sep 2021 - September 2026(expected)*
Doctor of Philosophy, Industrial Engineering and Operations Research. Advisor: Prof. Shipra Agrawal. *Proposed thesis title:* On theory and applications of online reinforcement learning.

University of Illinois, Urbana-Champaign, USA *Aug 2019 - Aug 2021*
Master of Science, Computer Science. Advisor: Prof. Nan Jiang.

Indian Institute of Technology, Kharagpur, India *July 2011 - June 2016*
Bachelor and Master of Technology in Electronics and Electrical Communication Engineering.

PROFESSIONAL EXPERIENCE

Part-time Research Scientist, Machgen AI [9 million seed] *June 2026 - Present*

- Leading design and development of smart model routing strategies for fast and optimized model inference to balance cost, latency, and quality of inference for image diffusion models.

Student Researcher, Meta Inc. (via Magnit, LLC) *November 2025 - April 2026*

- Design and analysis of a new post-training algorithm (RLVR): OC-GRPO that leverages privileged information at training time to improve deployment performance. **Keywords:** math-reasoning benchmarks; Self-guided curriculum; “learning cliff” in post-training.
- Focus on up to 14B parameter language models on a multi-GPU setup- [arxiv link](#).

Senior Analyst, Goldman Sachs India. *Jun 2016 - Aug 2018*

- Built high-throughput, low-latency financial risk calculation software; led risk modeling and monitoring.

TECHNICAL SKILLS

- Languages: Python, MATLAB, C++ .
- ML/DL Stack: PyTorch, HuggingFace Transformers, TRL, vLLM, FSDP.

- Infrastructure: AWS (multi-GPU clusters), SLURM, Weights & Biases.
- Techniques: GRPO, SFT, LoRA, reward modeling, evaluation pipeline design.

SELECTED AWARDS AND FELLOWSHIP

- Tang Family Fellow (awarded Fall 2021 by Columbia Engineering),
- Cheung-Kong Innovation Fellowship (awarded Fall 2024 by Cheung Kong Innovation Institute and Columbia Engineering).

WORKING PAPERS

Post-training algorithms for long horizon tasks

Accelerating useful credit assignment for long-horizon tasks with a sparse reward signal and limited environment access.

Off-Context GRPO: Learning to Reason on Hard Problems using Privileged Information Priyank Agrawal, Ankur Samanta, Shervin Ghasemlou, Jalaj Bhandari, Kavosh Asadi, Daniel Jiang, and Aditya Modi arXiv:2607.19313.

PAPERS

JOURNALS

Learning-Augmented Mechanism Design: Leveraging Predictions for Facility Location Priyank Agrawal, Eric Balkanski, Vasilis Gkatzelis, Tingting Ou, and Xizhi Tan, Mathematics of Operations Research (MOR) 2023 (subsumes the EC 2022 paper).

A Tractable Online Learning Algorithm for the Multinomial Logit Contextual Bandit Priyank Agrawal, Theja Tulabandhula, and Vashist Avadhanula, in European Journal of Operational Research (EJOR) 2023.

CONFERENCES

Q-learning with Posterior Sampling Priyank Agrawal, Shipra Agrawal and Azmat Azati. arXiv:2506.00917 ICLR 2026.

On the Convergence of Single-Timescale Actor-Critic Navdeep Kumar, Priyank Agrawal, Giorgia Ramponi, Kfir Yehuda Levy and Shie Mannor arXiv:2410.08868 NeurIPS 2025.

Optimistic Q-learning for average reward and episodic reinforcement learning Priyank Agrawal and Shipra Agrawal. arXiv:2407.13743 Conference on Learning Theory 2025, Journal version underway.

Policy Gradient with Tree Search (PGTS) in Reinforcement Learning Evades Local Maxima Navdeep Kumar, Priyank Agrawal, Kfir Yehuda Levy and Shie Mannor, in The Twelfth International Conference on Learning Representations-Tiny Paper track.

Learning-Augmented Mechanism Design: Leveraging Predictions for Facility Location Priyank Agrawal, Eric Balkanski, Vasilis Gkatzelis, Tingting Ou, and Xizhi Tan, The 23rd ACM Conference on Economics and Computation (EC 2022) (subsumed by MOR 2023 paper).

[C] **Improved Worst-Case Regret Bounds for Randomized Least-Squares Value Iteration** Priyank Agrawal, Jinglin Chen and Nan Jiang, in The 35th AAAI Conference on Artificial Intelligence (AAAI 2021).

Learning by Repetition: Stochastic Multi-armed Bandits under Priming Effect Priyank Agrawal and Theja Tulabandhula, in The 36th Conference on Uncertainty in Artificial Intelligence (UAI 2020).

Incentivising Exploration and Recommendations for Contextual Bandits with Payments Priyank Agrawal and Theja Tulabandhula, in the 17th European Conference on Multi-Agent Systems (EUMAS 2020).

ACADEMIC SERVICE

Teaching Assistant: At Columbia University IEOR4524 “Analytics in practice” and IEOR6614 “Optimization II”. At the University of Illinois Urbana-Champaign, CS101 “Introduction to programming for engineers and scientists” and CS446 “Machine learning”.

Reviewing: 30+ Operations Research Journal, ICLR 2025,2026; Book Review(Now Publisher) 2024, NeurIPS 2024,2025 AISTATS 2021,2024,2025.